

Lecture 03

Introduction - Part 02

Cyber Security and Privacy | NPTEL 106106248

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Video: <https://www.youtube.com/watch?v=nadHKp3egDY>

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The matter of concern is this, suppose you are riding a digital car and you are going at high speed, and you know that the even the speed is controlled by computer. And today, you have internet connected cars, you know, how does the car get updates? It is through the Internet. So, Tesla says when you wake up in the morning and when you enter the car, the car may be updated, as compared to what it was in the previous day.

So, the car becomes newer and newer every day- good, great, very exciting that is what the digital world is. But suppose, somebody gets access to your car, who is a bad guy or who you may term as hacker and if a hacker takes control of your car, when you are running at a high speed, just imagine, not just one car but the whole traffic. And these are potential scenarios for the future in transportation and that is where actually this report shows the International Centre for Automotive Technology, which actually test cars before they are released. Now, the test of cyber security has become much more stringent.

So and that is what this particular report says, every car is tested for cyber security and that is because of the potential damage of very, very serious nature which can cause damage to not just the car but damage to human life. So digital world can actually enable human life, it can also destroy human life today. It is not just about a computer attack where you lost some data from a computer but beyond that, it can actually bring systems to a halt and also damage people. Look at that, so this is September 16, 2019- this became a talk of the town, I remember when I was having my evening walk, faculty members particularly working in cyber security or in computer science were stopping and talking about this particular attack that happened on Ramco refinery in Saudi Arabia and this happened at 4 am in the morning and the company was going public and suddenly the news breaks that the refinery got shut down on this particular day, it got shut down not because it was short of raw materials, not because the workers did not report, not because there was a power failure, not because there was anything that is commonly understood as the cause of shutdown, but it was a cyber attack.

It was a cyber attack, there was a cyber attack and a refinery was halted on one fine morning, okay, and you see the tool used by the hackers, they used drones, okay, they use drones, okay they use drones to shut down a refinery, you understand the seriousness of this condition okay, of course nothing can stop IIT Madras, okay, so even if power goes off or even if every digital support system goes off, I can still continue to teach but you know that by and large several segments of the industry and government, are overly dependent on computers to function, okay, for day to day operations, the operational dependency of organizations especially with ERP systems, okay. You know, Citibank, long ago said we are a branchless bank, okay, what that means is we do not have we do not need branch, you know, it runs on

computers, it runs on network computers and ERP systems okay and if therefore the software stops functioning or the computer network stops functioning, the bank stops functioning, okay, even if people are present they cannot do anything, okay. So the criticality of information systems to run several organizations is very very high in the modern world, not just computer systems but the need to have internet to run organizations, to have connectivity to run organizations has become very high and very critical today and that is what you see in the Saudi oil refinery and today we have a development known as smart connected devices, smart connected devices. IOT is a example- Internet of Things, okay and you know in Internet of Things, there are devices other than, you know, the personal computers or the laptops which you use, of course they are computers but IOTs are like computers with processing units which can actually connect to the internet and transmit data, okay.

So for example, a sensor in a refinery, a temperature sensor, okay, in a refinery which is nothing but a device which may not have any display but it is a sensor, it may be cable connected or it may be cable less but it is actually sensing and transmitting data about temperature okay and since it is internet connected, today experts say any device that is connected to internet is not safe, that is the basic principle, any device, any system that is internet connected is not safe, okay. So if an IOT is connected to internet and it is sensing a very important parameter for process control and if it comes under the control or if it is hacked or some unauthorized access happens, you can imagine what happens, okay, if temperature is, say 500 degree centigrade and the hacker makes it say 5 degree centigrade, you know the controls that comes into picture you know, it actually can totally damage the system, okay, in terms of because of the wrong signal. So, IOTs can do a lot of good - Manufacturing 4.0, okay, it is based on, you know, this high end devices or smart connected devices, okay, and you see a lot of hype in literature or trade magazines that 4.0, you know we are going to the new generation of digital technologies and manufacturing with digital technologies but that is a bright side, okay, this can do a lot of good things. When I used to work in manufacturing, 10 years in process control and we did not have much of digital technologies, it was mostly analog technologies and of course computers were used in control but if there is a sensor, the sensor is connected by wire to a controller okay, so there is a signal that actually, standard signal that transmit between the sensor and the controller, okay and therefore there is no way somebody can access that and manipulate that etc.

But the same sensor when it becomes digital and digital plus internet connected, okay, so the potential for unauthorized access becomes high because it is internet connected and when the whole world relies on digital technologies connected to internet, instances like this actually becomes eye openers. So, it brings in the severity of the potential dark world that can actually make all this extremely vulnerable, you know, the world is vulnerable and look at this, of course there was politics in this after the attack happened and the US said Iraq Iran is behind the Saudi attacks and then you see what happened to Iran a few months later. A US drone strike on Baghdad airport which killed Qasem Soleimani who was actually a dear citizen or the commander and highly respected name for people of Iran, okay so that is a country to which he belonged, of course a different country can look at the same person from a different perspective but to that country he was a great hero and he was killed and who killed him?

That's the important question, it is not a soldier who killed him, okay, it is not a gunshot but what killed him was a drone, it was a drone that was used by the United States to kill a military leader.

So drones have come, drones have come into the picture of the cyber world and that is the important point I want to bring to your notice, when as we start this course on cyber security, okay, cyber security used to be called as information security or there are two courses or two topics information security and cyber security, okay. So cyber security is a more recent term that is used to describe security of computer systems in the modern world but generally people believed what is most important in computer systems is the data, because in information systems you must have learned, what is the purpose of information systems? Basically information, they are information systems, okay, they create data, they store data, they transmit data, they process data, okay and they present data, okay, data to information, to knowledge and so on, okay, in addition to process automation. So that is what computer systems do or that is what information systems do but you see the scope widening in recent times from security to information versus security to infrastructure, security to people, what has become more insecure is not just machines, not just data but also people, the potential to cause damage to people has become real, it is not imaginary, it is not like a scientific fiction, okay, it is not more a fiction, this is as actually happened and here is more from my collection of questions of course, you must have heard about Air India's data breach few years ago and how it actually caused huge embarrassment to the company and here is another technology company, which some of you may be members of and which is very much in the news, the Twitter, okay A technology company coming under cyber attack, okay and you do not expect that to happen right, they we generally expect them to be much smarter than the hackers but the Twitter handles of celebrities like Bill Gates or the contestant for American elections was hacked by hackers and that was a huge embarrassment to Twitter at some point in time and they had to explain, why it happened and what actions they are taking, they had to come in public and give an explanation, okay, and this is something that you can imagine when you are a reputed organization or you are working for a technology organization, okay, suppose you are working for TCS, the Indian company, okay, you know that TCS website was hacked some years ago and that was a joke because every newspaper reported it in as a highlight, okay or as a headline, because it is a fun reporting it, okay, an IT company's website hacked, okay and that is all they know about technology, you can actually make fun of them okay and TCS had to explain that, well, we do not maintain our own website, it is outsourced, okay.

Alright, so here is the summary, so I am not going to teach cyber incidents and why it happened and get into politics or get into popular media much but the purpose was to give you an overview of cyber security in the current world, okay and the first thing I want to summarize is cyber security is a current and serious problem, okay, it is happening okay, more digital would invite more cyber security problems, okay, when digital computers were less or computerization as we called it was less, we did not have much news about cyber attack in the newspapers. If you read newspapers of the 60s or the 50s, nobody talks about cyber attack, nobody about talks about information security, you know, probably there was no course like

that, okay but with more adoption of technology, we have the growing concern about protecting this technology, okay, against the evil, okay, against the dark world, okay and popular media suggest to us this and the second reason as I said, is the pervasiveness of digital technologies and all of us today use digital technology on a daily basis, what is the frequency of you looking at your phone? how many times you look at your phone? you need frequent updates right, you keeps running in you, I am actually missing my phone, I have kept it from me because I want to focus on you but the phone is there this is something that you want to take you know, so some scholars say frequent information is a need for people.

So when you are getting it you know, you use it does not mean it is addiction, does not necessarily mean it is negative. Although elders may scold you that you are closer to the phone than to us or to me etc but that is norm of the day, okay and I do not see it changing okay, I do not see the world walking back or turning the clock back to the 50s or 60s, we are going to go forward, okay, but what that essentially means is that we have to manage, we have to learn to manage the cyber threats with which we live, okay The cars can run into accidents. So, solution one -stop using cars, right and solution two- increase safety -focus on safety so I think route two is sensible, right, we cannot stop using digital technologies we cannot throw away our phones and today live, okay or we cannot stop using the internet, okay, some people do, okay, there I, I know people, IT professionals are, not IT professionals, IT faculty who do not use instant messaging like whatsapp they just do not want to use, okay. So, you can keep away, there there are people who do not use televisions at home, there are people who do not have any social media account, so that is one way of protecting yourself, of course, to protect yourself you lose something, right, to gain something, you lose something so it depends on what you want to gain but generally we see people are not willing to lose the privileges of the digital world, okay, the the better life that it gives us or it promises us, okay.

Autonomous cars may be great future of transportation when, when transportation becomes much more efficient, okay and also it becomes personalized all that is huge potential for future, but we should also be aware that it all comes at a cost and safety is something that should not be compromised. So, essentially it only tells us that attention- attention due attention should be paid to the the threats or the cyber security at different levels. The second point I highlight there is, cyber security affects different units okay, it affects individuals, okay, you can recall some of the instances I reported which is an individual level attack, you may not, it is not because you are an employee that you are getting attacked, it is not an organizational infrastructure assets that is being attacked but it's attack is on you, okay, your bank account or your individual data. So individuals, organizations, society and government are all vulnerable, okay, these are all different units, so all these units have cyber challenges or cyber security challenges, okay so the the landscape is wide, it affects different domains, it affects different types of organizations, we talked about healthcare versus manufacturing, All these have different implications So it is not restricted to certain domain alone, it is it is quite open to different domains, okay and government should also be concerned about cyber security in two ways number one, government systems can be attacked, government data can be leaked, okay, government possesses a lot of data, you agree with it, with Aadhaar which will be one of our case studies, as we proceed in the course, okay, it is a load of personal data

that government collects and stores and once it goes out to the wrong hands, okay, it is people's data or people's privacy, that is compromised, okay, so there is a cost to it.

So government should be concerned about cyber security and also government's role is to ensure the safety of citizens, the welfare of citizens, essentially the welfare from that point of view its government's role also to formulate policies and regulations, regulate the cyber world in such a way, that it actually, the the country is safe and the country progress and the country does tap into the digital technologies without compromising security, okay, so that is a big order, okay and that is the challenge every government is facing, okay. No government wants to discontinue digital technologies, particularly India, you know so the government is tech savvy, which is good, okay, but at the same time there are huge challenges that are related to privacy and data protection, okay, privacy issues and data protection issues which is a close cousin of cyber security, okay, so both are interrelated, okay, it is getting national attention, okay and when we explore the regulation across the globe on data protection, we will see, of course the developed countries and also a developing country like India, okay, what is happening in data protection world in India, okay and of course this is related to technology, this is related to privacy and it is also related to politics and I must preempt potential biases that you may see in me when I discuss politics, okay, so because each of us can have an opinion in politics, okay, but I would try to remain as government political party neutral as much as possible but different governments face different challenges, okay and they change their taunts when they move from opposition to ruling party but privacy is an important issue of the day, okay. And the fourth point as a background and motivation for the course I want to stress is the role of technology in cyber security. I would articulate that technology has a triple role, okay, technology plays three roles in cyber security, technology can be source of threat, okay, so somebody talked about denial of service attack okay, denial of service attack is actually an attack on a network or a computer using another computer. It is a script that runs on one computer to sort of stall another computer, so it is a technology that is used for cyber attack, okay, so technology itself is a source of threat.

That is one role of technology, okay and there are scripts available in public, if you want to actually try out denial of service attack, okay. Second, it is also an asset to be protected, it can be a source of threat, it is also the asset that organizations need to protect, okay. When your data center, your databases, okay or your important devices like your computer - in ransomware your computer is the asset that actually gets attacked. That is the asset that you need to protect, okay, so technology is a threat, technology is an asset and technology can also be used as a defense weapon, okay, so how do you defend your technology? So protection mechanisms which we will discuss in one of the sessions, predominantly uses technology it deploys technology, the firewall technology for example, is a technology to defend or protect your assets okay and therefore my my articulation that technology is triple, okay, it is not, it is not just one it can be a source of threat, it can be a weapon and it is also the asset that needs protection So when we use the term technology in cyber security, keep in mind what are you referring to okay, and that clear understanding is important.

Generally we may think that, well cyber security technologies mean technologies that you use to protect your assets, no not necessary, okay, not necessarily you need to understand the

nuances.